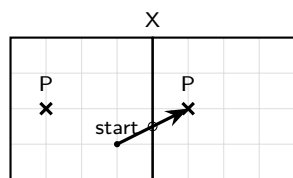
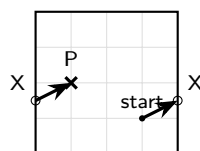


A shot travels straight. At an X edge it reappears at the same height on the other X edge. Y edges work the same way, at the same left-to-right position. It keeps its direction. All four corners are the same S. Stop at the first T reached.

A shield is one exact point, away from S and T. A shot stops only if its line passes through that point. A counter marks the point at its center. On an unfolded map, a shield repeats at the same place in every large room-copy.

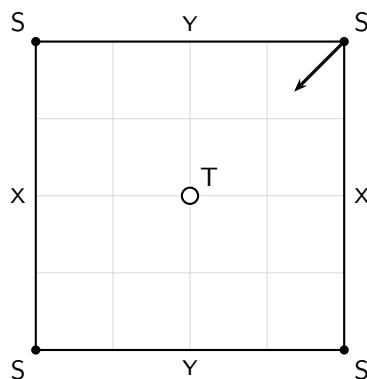
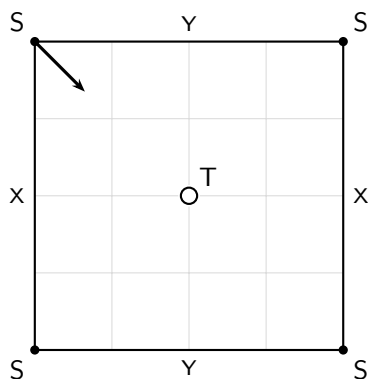
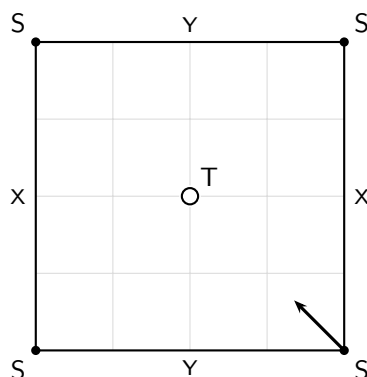
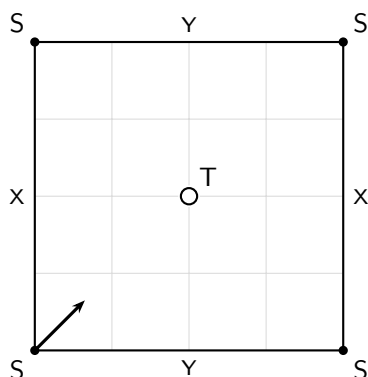


one shield P in two copies



the same shot in one room

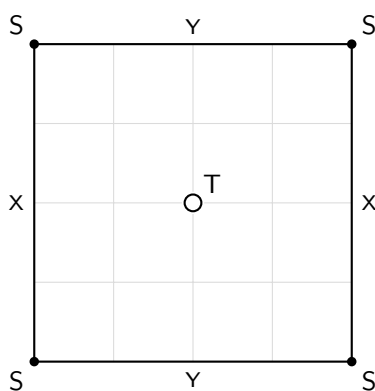
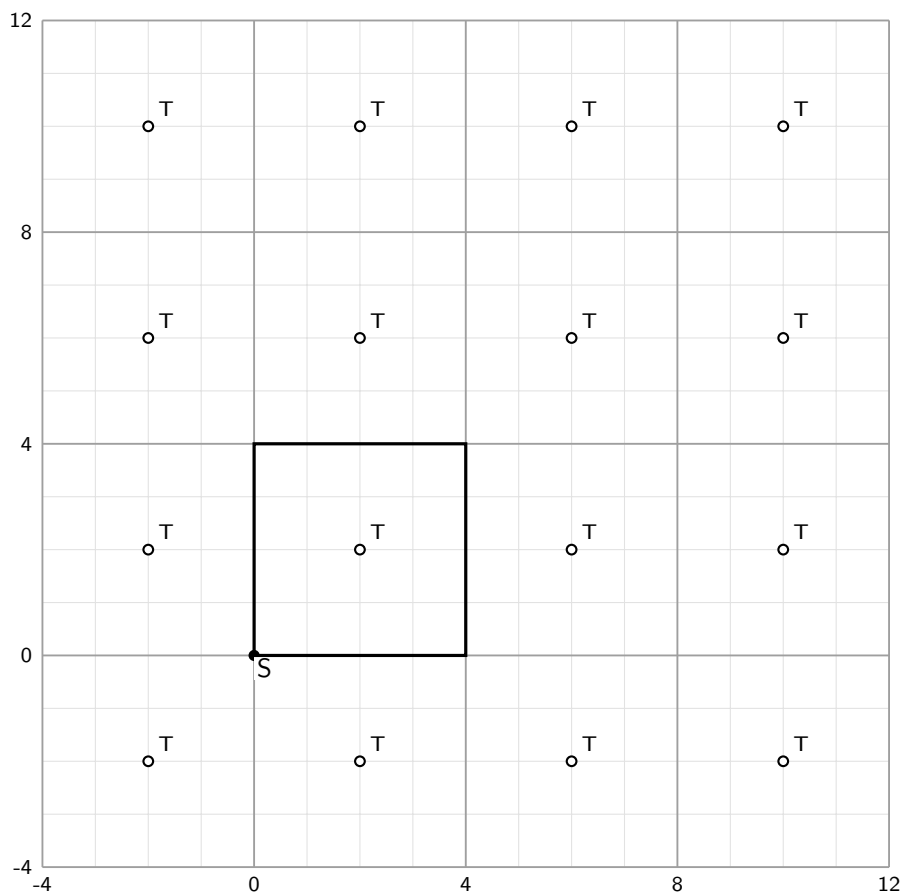
Problem 1: Finish the four diagonal shots. Choose one shield arrangement for the room that stops all four shots. Use as few shields as you can. Your partner checks the shot lines and shields. Swap jobs for a new arrangement.



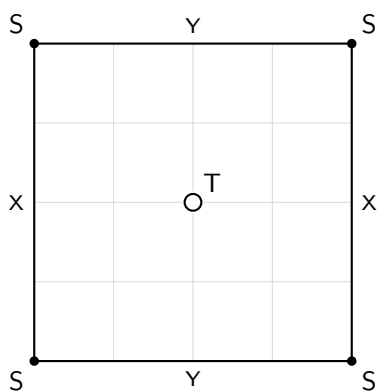
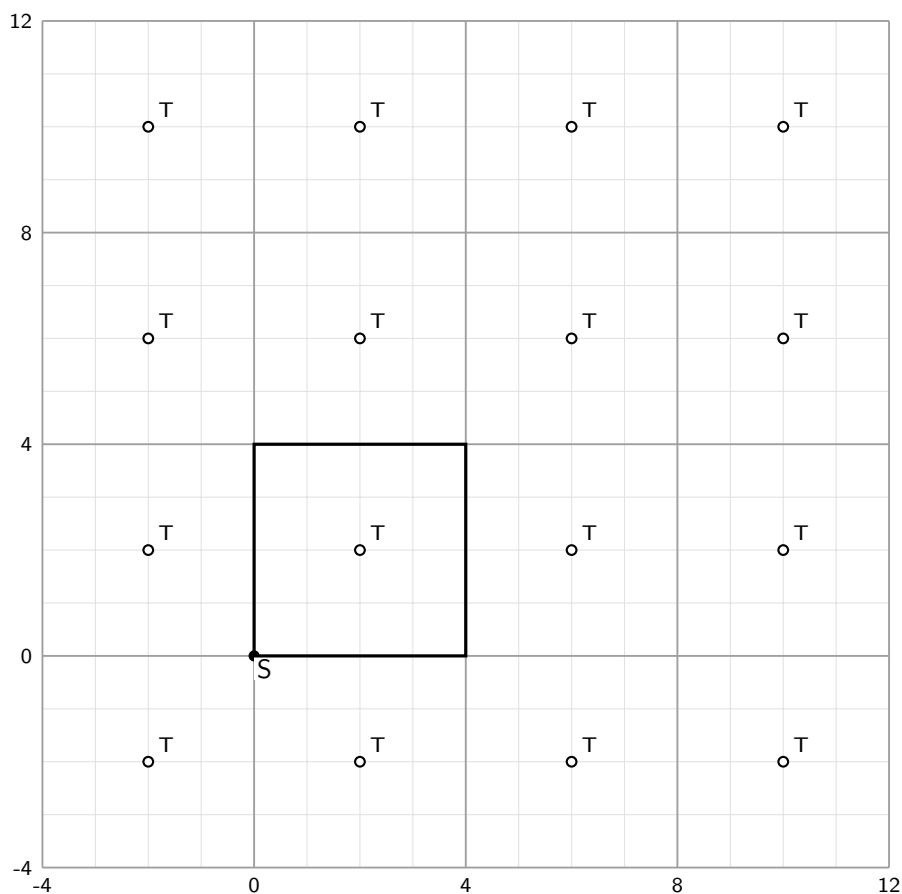
Our fewest shields: _____

Problem 2: Aim straight shots from S at the T copies at (2, 6), (6, 2), (6, 6) and (10, 6). Your partner marks the first T each shot reaches. In the small room, draw one shot that crosses a portal before reaching T.

Each large square below is another copy of the same room. Every T is the same target. The bold square is the room shown below the map.



Problem 3: Find four shield points in the room that stop every shot on the map. Your partner chooses targets and uses straight shots on the map to test your shields. Move your shields until no shot gets through.



Problem 4: Find four shield points that stop every straight shot from S to T, even one that goes far beyond the map. Explain why your points work. Could three shields ever stop every shot?

